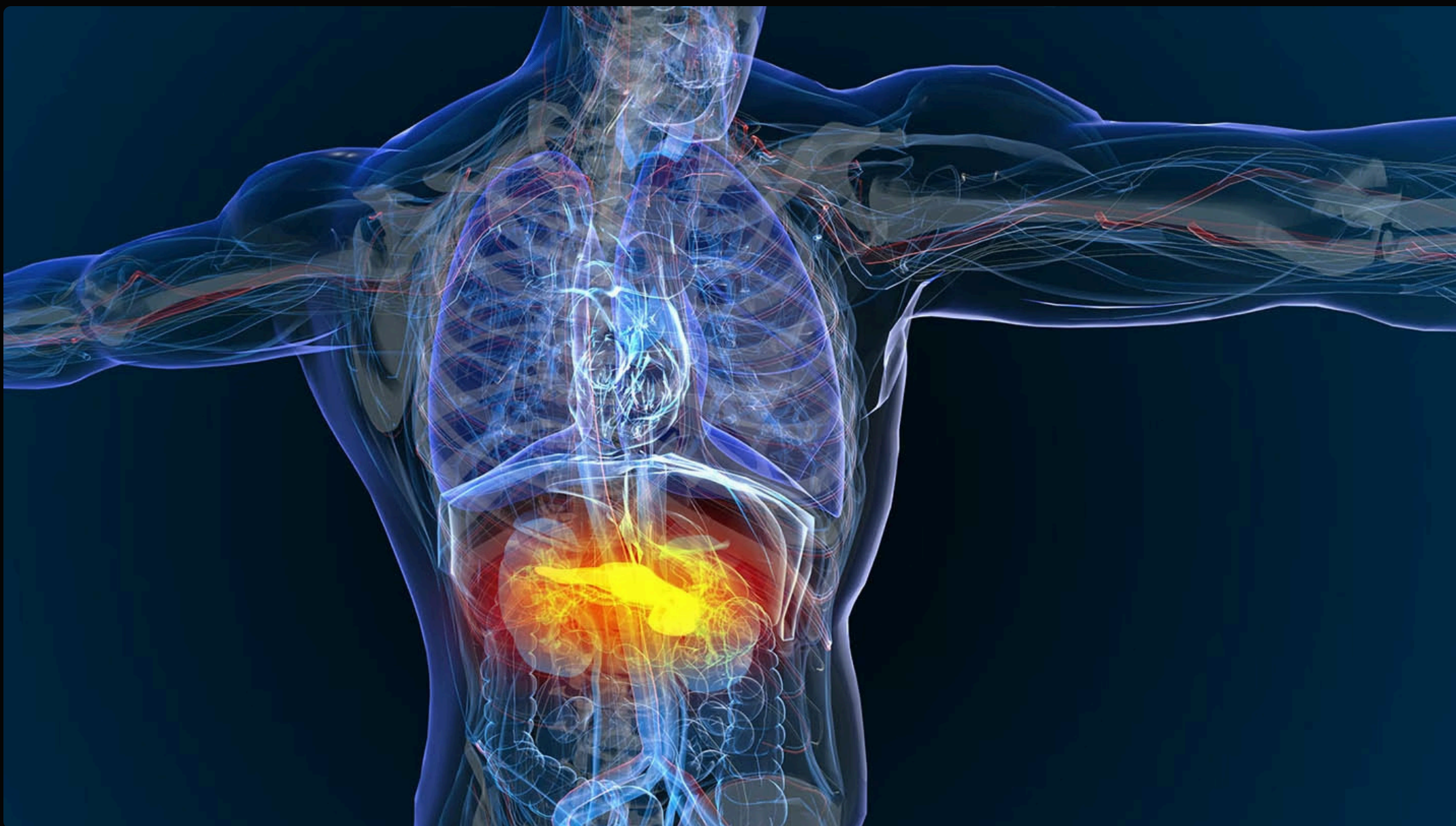




AI Detects Pancreatic Cancer Before Radiologists Can

A breakthrough AI framework called **REDMOD** (Radiomics-based Early Detection Model) has demonstrated the ability to identify Stage 0 pancreatic ductal adenocarcinoma (PDAC) up to **3 years before clinical diagnosis** - outperforming expert radiologists and offering real hope for one of oncology's most difficult cancers. Published in *Gut* (May 2026), this research from Mayo Clinic represents a potential paradigm shift from late-stage symptomatic diagnosis to proactive preclinical interception.

Why Pancreatic Cancer Is So Hard to Catch Early

Pancreatic ductal adenocarcinoma is notorious for its **late-stage presentation**. By the time symptoms appear - jaundice, weight loss, abdominal pain - the disease has typically advanced beyond surgical resectability. The 5-year survival rate for PDAC remains below 13%, largely because the vast majority of cases are diagnosed at Stage III or IV.

Stage 0 PDAC, also called **carcinoma in situ**, is **visually occult** on standard CT imaging - meaning tumors are present but not yet visible to the naked eye. Radiologists reviewing these scans appear normal. This is precisely the window where AI has the potential to intervene.

⚠ The median survival for metastatic PDAC is less than 12 months. Early detection at Stage 0 could dramatically change that trajectory.

THE CHALLENGE

The Diagnostic Gap REDMOD Was Built to Close

Visually Occult Tumors

Stage 0 PDAC does not produce visible masses on CT scans. Human radiologists have no reliable visual cue to act on, leading to near-universal missed diagnoses at this stage.

Subtle Tissue Changes

Even before a tumor is visible, the pancreatic tissue undergoes microscopic structural and textural changes - changes that are imperceptible to the human eye but detectable through advanced quantitative imaging analysis.

No Scalable Screening Tool

Unlike breast or colon cancer, there is no established population-level screening protocol for PDAC. High-risk patients (those with genetic mutations, family history, or new-onset diabetes) have no reliable triage mechanism.

How REDMOD Works: The Methodology Explained

REDMOD combines two powerful AI techniques into a single automated pipeline - designed to find invisible signals in CT scans that even expert radiologists miss.

Radiomic Feature Extraction

Extract 968 quantitative imaging features.

Feature Selection

Reduce features to 40 via mRMR selection.

CT Input & Segmentation

Volumetric pancreas segmented with 3D nnU-Net.

Ensemble Modeling

Combine LR, RF, and XGBoost to score cancer likelihood.

The fully automated nature of this pipeline means no manual intervention is required between scan acquisition and risk output - making it inherently scalable for real-world clinical deployment.

Deep Dive: Two Core Technologies Behind REDMOD



Deep Learning Segmentation (3D nnU-Net)

The first stage of REDMOD uses a **3D nnU-Net**, a deep learning architecture specifically designed for volumetric medical image segmentation. It precisely outlines the pancreas in 3D space across the entire CT scan - a task that is time-consuming and variable when done manually by radiologists.

This automated segmentation ensures that subsequent analysis is focused exactly on the right tissue, removing human variability and enabling consistent, reproducible results across different institutions and CT scanners.



Radiomics: Reading What Eyes Can't See

Radiomics is the science of extracting thousands of quantitative features from medical images - texture, shape, intensity patterns, and spatial relationships - that are imperceptible to human observers.

REDMOD initially extracts **968 features** per scan. Using a mathematical selection process (minimum redundancy maximum relevance), these are distilled down to **40 high-signal features**, 90% of which came from **wavelet-filtered images** - a mathematical transformation that reveals hidden textural patterns not visible in raw scans.

Why an Ensemble Model? The Logic Behind the Design

Rather than relying on a single algorithm, REDMOD uses a **heterogeneous ensemble** - three different machine learning models working together. This approach was deliberately chosen because no single algorithm performs optimally across all data types and clinical scenarios.



Logistic Regression

A classical statistical model that provides interpretable probability scores. It excels at identifying linear relationships between features and cancer presence, offering transparency and clinical interpretability.



Random Forest

An ensemble of decision trees that captures non-linear feature interactions. It is robust to noise and overfitting, and handles the high-dimensional radiomic feature space effectively.



Extreme Gradient Boosting (XGBoost)

A state-of-the-art boosting algorithm that iteratively corrects prediction errors. It is particularly powerful for structured tabular data and has become a benchmark in medical ML competitions.

Combining all three into a single ensemble reduces the risk that any one model's weaknesses affect the final prediction - improving both sensitivity and robustness across diverse patient populations.

Study Design: How the Evidence Was Built



A Large, Multi-Institutional Dataset

The study included **1,462 CT scans** gathered from multiple institutions and CT vendors - intentionally designed to reflect real-world clinical variability rather than idealized conditions.

- **219 prediagnostic scans** from patients later diagnosed with PDAC (median lead time: 427 days; range: 90–1,092 days)
- **1,243 control scans** from cancer-free individuals confirmed with ≥ 3 years of follow-up
- Split into a **training cohort** ($n = 969$) and an **independent test cohort** ($n = 493$)

The independent test cohort was used to benchmark REDMOD against two board-certified abdominal radiologists in a head-to-head multireader study - each radiologist rating the likelihood of Stage 0 PDAC on a 1–5 scale.

Key Results: REDMOD vs. Radiologists

The performance gap between REDMOD and expert radiologists was striking - particularly at the longest lead times, where early interception matters most.

73%

REDMOD Sensitivity

Detected Stage 0 PDAC in prediagnostic scans vs. only 38.9% pooled sensitivity for radiologists - nearly double the detection rate.

0.82

AUC Score

Area under the curve in the independent test cohort (95% CI: 0.81-0.83), demonstrating strong discriminatory power across diverse imaging platforms.

475

Median Lead Days

REDMOD detected cancers at a median of 475 days before clinical diagnosis - providing a critical window for intervention.

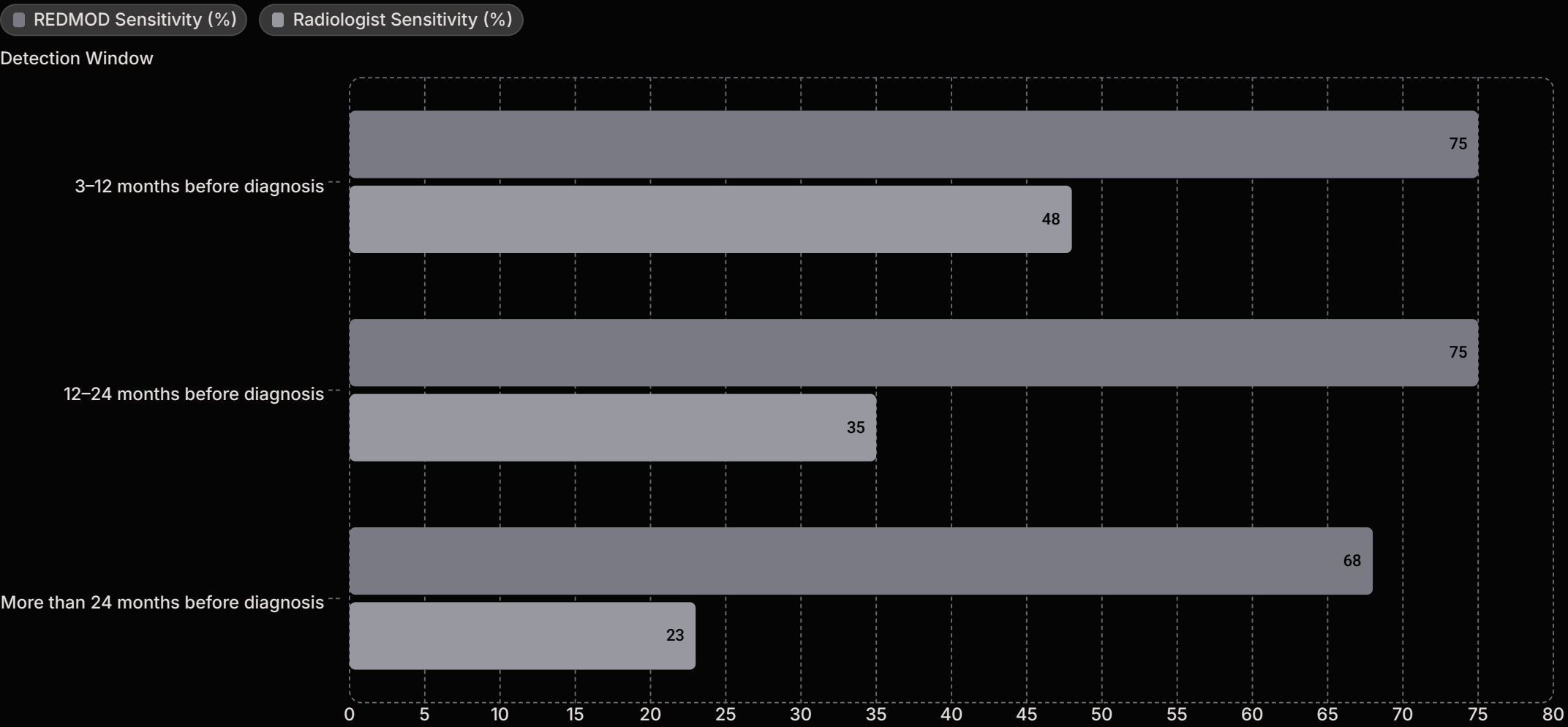
91%

Longitudinal Concordance

90-92% concordance on repeat imaging confirms the model produces stable, reproducible results over time - essential for clinical monitoring use.

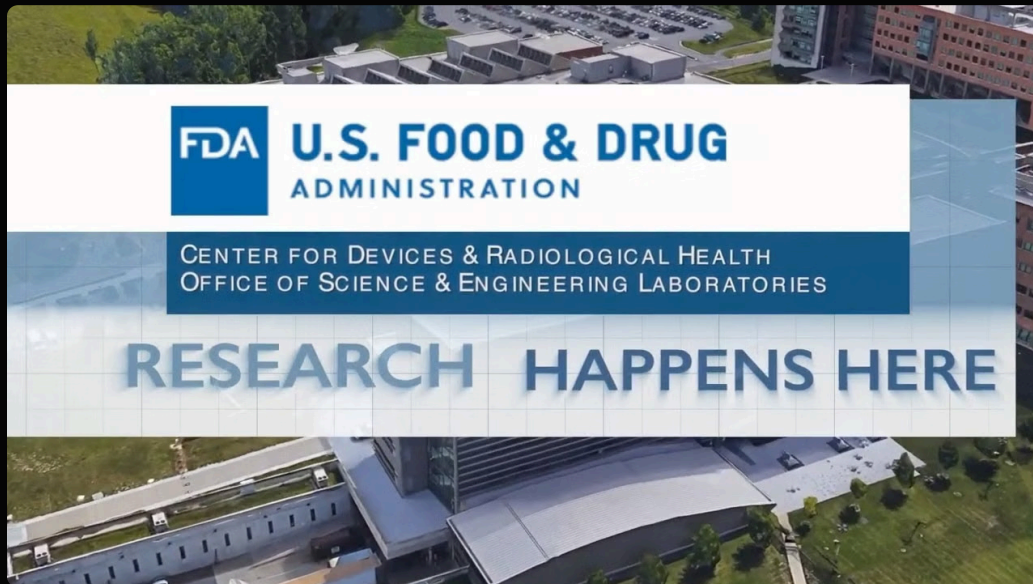
Performance Over Time: The Earlier, the Greater the Advantage

One of the most compelling findings is how REDMOD's advantage over radiologists **grows with longer lead times** - meaning the model is most valuable precisely when human eyes are least reliable.



At more than 24 months before diagnosis - the most clinically impactful window - REDMOD achieved **68% sensitivity** versus just **23% for radiologists**. This three-fold advantage underscores the model's unique ability to detect pre-tumor imaging signatures years before conventional methods.

FDA Parallel: AI Enters the Clinical Trial Pipeline



A Regulatory Agency Embraces AI

In parallel with advances like REDMOD, the **FDA announced a landmark pilot program** (May 2026) to use AI for optimizing early-phase clinical trials. The initiative addresses persistent challenges: inefficient dose escalation decisions, lengthy timelines, limited patient populations, and regulatory uncertainty.

Two proof-of-concept **real-time clinical trials** were initiated by AstraZeneca and Amgen, in partnership with **Paradigm Health** - an AI platform that integrates EHR data and transmits safety signals to both the FDA and trial sponsors in real time.

"For 60 years, we've been conducting clinical trials in the same way, where key data signals can take years to reach the FDA."

- **Marty Makary, MD, MPH, FDA Commissioner**

What This Means for the Future of Oncology

Non-Invasive Triage for High-Risk Patients

REDMOD could serve as a first-line screening tool for individuals with known PDAC risk factors - including BRCA2 mutations, familial pancreatic cancer history, or new-onset diabetes - enabling targeted use of more invasive confirmatory workups.

Longitudinal Monitoring Tool

With 90-92% concordance on repeat imaging, REDMOD is well-suited for serial surveillance. Patients flagged as high-risk could be monitored over time, with the model tracking subtle progression signals before a tumor becomes visible.

A Template for Other Cancers

The radiomics + deep learning ensemble methodology is not unique to PDAC. The REDMOD framework's architecture could be adapted for other cancers with similarly challenging early detection profiles - including ovarian, lung, and hepatocellular carcinomas.

Key Takeaways & Next Steps

01

Prospective Validation Is the Next Priority

REDMOD must be validated in prospective, high-risk cohort studies before clinical deployment. Retrospective performance, while exceptional, must be confirmed in real-world screening workflows.

- ✓ REDMOD represents a foundational step toward making Stage 0 pancreatic cancer detection a clinical reality - moving AI from research benchmark to life-saving screening tool.

02

Multi-Institutional Expansion

The study already included multiple institutions and CT vendors - but broader global validation across diverse racial, ethnic, and geographic populations is essential for generalizability.

- i Funded by the NIH, Mayo Clinic Comprehensive Cancer Center, Funk Zitiello Foundation, Centene Charitable Foundation, and the Hoveida Family Foundation. Published in *Gut*, April 2026. Senior author: Ajit Harishkumar Goenka, MD, Mayo Clinic.

03

Integration with Clinical Workflow

For REDMOD to shift clinical practice, it must integrate seamlessly into existing radiology workflows as an automated background process - flagging high-risk scans for follow-up without disrupting routine reads.

04

Regulatory Pathway

FDA clearance as a clinical decision support tool will require additional prospective data, bias assessments, and alignment with the FDA's AI Risk Management Framework - now actively being developed.